Switch-mode Series PNP Silicon Power Transistors

MJE5850, MJE5851, MJE5852

The MJE5850, MJE5851 and the MJE5852 transistors are designed for high−voltage, high−speed, power switching in inductive circuits where fall time is critical. They are particularly suited for line operated switch−mode applications.

Features

- Switching Regulators
- Inverters
- Solenoid and Relay Drivers
- Motor Controls
- Deflection Circuits
- Fast Turn−Off Times
- Operating Temperature Range −65 to +150°C
- 100°C Performance Specified for:
  - Reversed Biased SOA with Inductive Loads
  - Switching Times with Inductive Loads
  - Saturation Voltages
  - Leakage Currents
- Complementary to the MJE13007 Series
- These Devices are Pb−Free and are RoHS Compliant*

MAXIMUM RATINGS

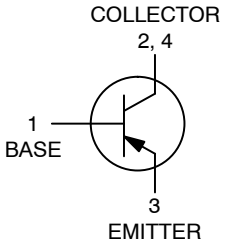
Rating	Symbol	Value	Unit
Collector−Emitter Voltage MJE5850 MJE5851 MJE5852	$V_{CEO(sus)}$	 300 350 400	Vdc
Collector−Emitter Voltage MJE5850 MJE5851 MJE5852	V_{CEV}	 350 400 450	Vdc
Emitter Base Voltage	V_{EB}	6.0	Vdc
Collector Current − Continuous (Note 1)	I_C	8.0	Adc
Collector Current − Peak (Note 1)	I_{CM}	16	Adc
Base Current − Continuous (Note 1)	I_B	4.0	Adc
Base Current − Peak (Note 1)	I_{BM}	8.0	Adc
Total Power Dissipation @ T_C = 25°C Derate above 25°C	P_D	 80 0.640	 W W/°C
Operating and Storage Junction Temperature Range	T_J, T_{stg}	−65 to 150	°C

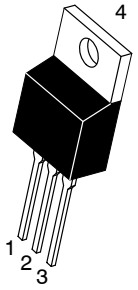
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. Pulse Test: Pulse Width = 5 ms, Duty Cycle ≤ 10%.

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

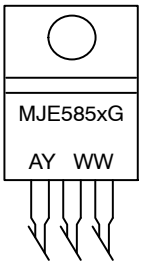
8 AMPERE
PCP SILICON
POWER TRANSISTORS
300−350−400 VOLTS
80 WATTS





TO−220
CASE 221A−09
STYLE 1

MARKING DIAGRAM



MJE585x	=	Device Code x = 0, 1, or 2
G	=	Pb−Free Package
A	=	Assembly Location
Y	=	Year
WW	=	Work Week

ORDERING INFORMATION

See detailed ordering and shipping information in the package dimensions section on page 7 of this data sheet.

NOTE: Some of the device on this data sheet have been **DISCONTINUED**. Please refer to the table on page 7

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THERMAL CHARACTERISTICS

Rating	Symbol	Max	Unit
Thermal Resistance, Junction−to−Case	$R_{\theta JC}$	1.25	°C/W
Maximum Lead Temperature for Soldering Purposes: 1/8″ from Case for 5 Seconds	T_L	275	°C

ELECTRICAL CHARACTERISTICS (T_C = 25°C unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Collector−Emitter Sustaining Voltage (I_C = 10 mA, I_B = 0) MJE5850 MJE5851 MJE5852	$V_{CEO(sus)}$	 300 350 400	 – – –	 – – –	Vdc
Collector Cutoff Current (V_{CEV} = Rated Value, $V_{BE(off)}$ = 1.5 Vdc) (V_{CEV} = Rated Value, $V_{BE(off)}$ = 1.5 Vdc, T_C = 100°C)	I_{CEV}	 – –	 – –	 0.5 2.5	mAdc
Collector Cutoff Current (V_{CE} = Rated V_{CEV}, R_{BE} = 50 Ω, T_C = 100°C)	I_{CER}	–	–	3.0	mAdc
Emitter Cutoff Current (V_{EB} = 6.0 Vdc, I_C = 0)	I_{EBO}	–	–	1.0	mAdc
SECOND BREAKDOWN					
Second Breakdown Collector Current with base forward biased	$I_{S/b}$	See Figure 12			
Clamped Inductive SOA with base reverse biased	RBSOA	See Figure 13			
ON CHARACTERISTICS (Note 2)					
DC Current Gain (I_C = 2.0 Adc, V_{CE} = 5 Vdc) (I_C = 5.0 Adc, V_{CE} = 5 Vdc)	h_{FE}	 15 5	 – –	 – –	–
Collector−Emitter Saturation Voltage (I_C = 4.0 Adc, I_B = 1.0 Adc) (I_C = 8.0 Adc, I_B = 3.0 Adc) (I_C = 4.0 Adc, I_B = 1.0 Adc, T_C = 100°C)	$V_{CE(sat)}$	 – – –	 – – –	 2.0 5.0 2.5	Vdc
Base−Emitter Saturation Voltage (I_C = 4.0 Adc, I_B = 1.0 Adc) (I_C = 4.0 Adc, I_B = 1.0 Adc, T_C = 100°C)	$V_{BE(sat)}$	 – –	 – –	 1.5 1.5	Vdc
DYNAMIC CHARACTERISTICS					
Output Capacitance (V_{CB} = 10 Vdc, I_E = 0, f_{test} = 1.0 kHz)	C_{ob}	–	270	–	pF
SWITCHING CHARACTERISTICS					
Resistive Load (Table 1)					
Delay Time	(V_{CC} = 250 Vdc, I_C = 4.0 A, I_{B1} = 1.0 A, t_p = 50 μs, Duty Cycle ≤ 2%)	t_d	–	0.025	0.1	μs
Rise Time		t_r	–	0.100	0.5	μs
Storage Time	(V_{CC} = 250 Vdc, I_C = 4.0 A, I_{B1} = 1.0 A, $V_{BE(off)}$ = 5 Vdc, t_p = 50 μs, Duty Cycle ≤ 2%)	t_s	–	0.60	2.0	μs
Fall Time		t_f	–	0.11	0.5	μs
Inductive Load, Clamped (Table 1)						
Storage Time	(I_{CM} = 4 A, V_{CEM} = 250 V, I_{B1} = 1.0 A, $V_{BE(off)}$ = 5 Vdc, T_C = 100°C)	t_{sv}	–	0.8	3.0	μs
Crossover Time		t_c	–	0.4	1.5	μs
Fall Time		t_{fi}	–	0.1	–	μs
Storage Time	(I_{CM} = 4 A, V_{CEM} = 250 V, I_{B1} = 1.0 A, $V_{BE(off)}$ = 5 Vdc, T_C = 25°C)	t_{sv}	–	0.5	–	μs
Crossover Time		t_c	–	0.125	–	μs
Fall Time		t_{fi}	–	0.1	–	μs

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

2. Pulse Test: PW = 300 μs. Duty Cycle ≤ 2%

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TYPICAL ELECTRICAL CHARACTERISTICS

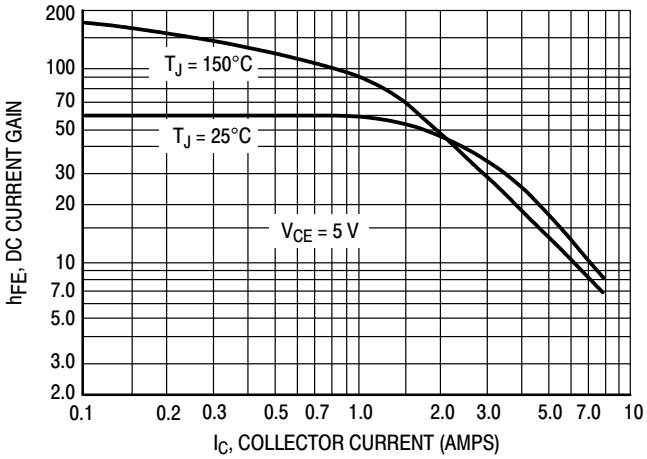


Figure 1. DC Current Gain

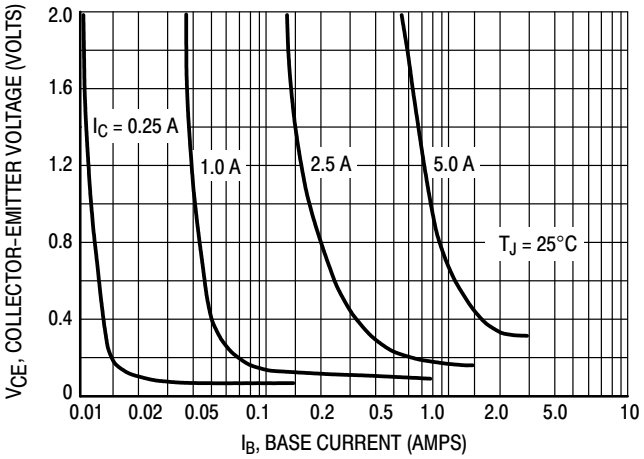


Figure 2. Collector Saturation Region

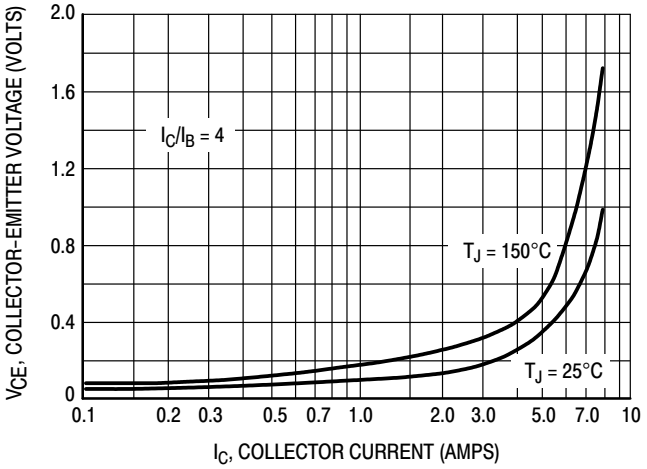


Figure 3. Collector−Emitter Saturation Voltage

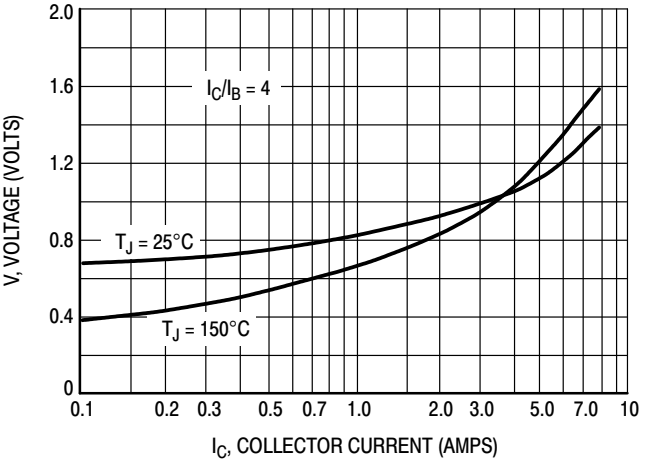


Figure 4. Base−Emitter Voltage

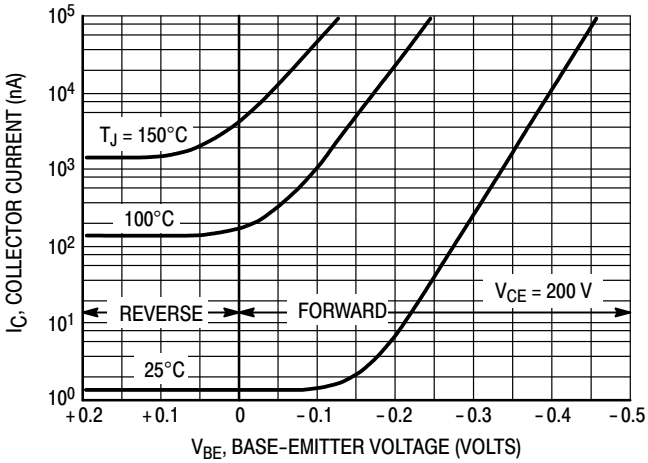


Figure 5. Collector Cutoff Region

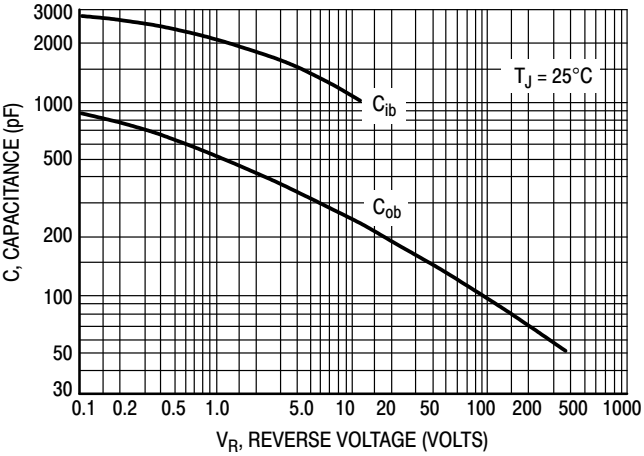
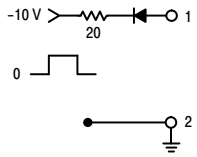
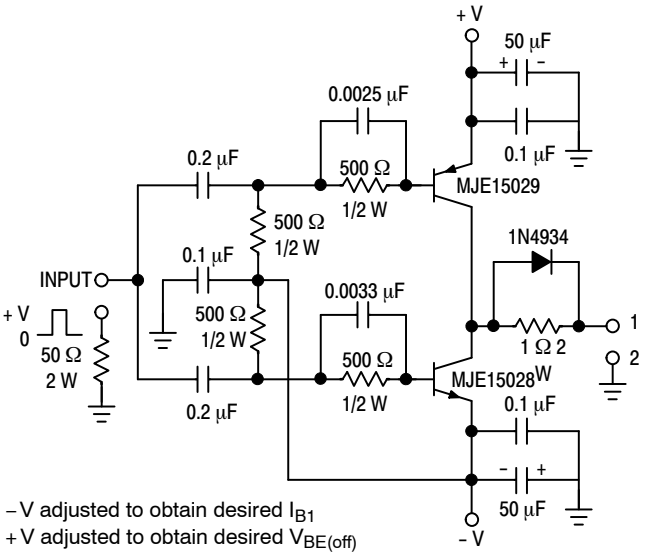
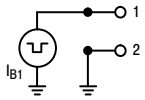
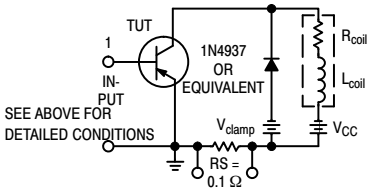
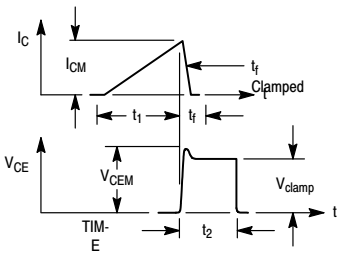
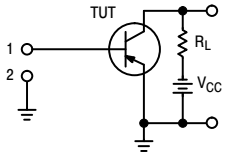


Figure 6. Capacitance

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Table 1. TEST CONDITIONS FOR DYNAMIC PERFORMANCE

	$V_{CEO(sus)}$	RBSOA AND INDUCTIVE SWITCHING	RESISTIVE SWITCHING
INPUT CONDITIONS	 PW Varied to Attain I_C = 100 mA		 I_{B1} adjusted to obtain the forced h_{FE} desired **TURN–OFF TIME** Use inductive switching driver as the input to the resistive test circuit.
CIRCUIT VALUES	L_{coil} = 80 mH, V_{CC} = 10 V R_{coil} = 0.7 Ω	L_{coil} = 180 μH R_{coil} = 0.05 Ω V_{CC} = 20 V V_{clamp} = 250 V R_B adjusted to attain desired I_{B1}	V_{CC} = 250 V R_L = 62 Ω Pulse Width = 10 μs
TEST CIRCUITS	**INDUCTIVE TEST CIRCUIT**  SEE ABOVE FOR DETAILED CONDITIONS	**OUTPUT WAVEFORMS**  t_1 Adjusted to Obtain I_C $t_1 \approx \frac{L_{coil}(I_{CM})}{V_{CC}}$ $t_2 \approx \frac{L_{coil}(I_{CM})}{V_{Clamp}}$ Test Equipment Scope — Tektronix 475 or Equivalent	**RESISTIVE TEST CIRCUIT** 

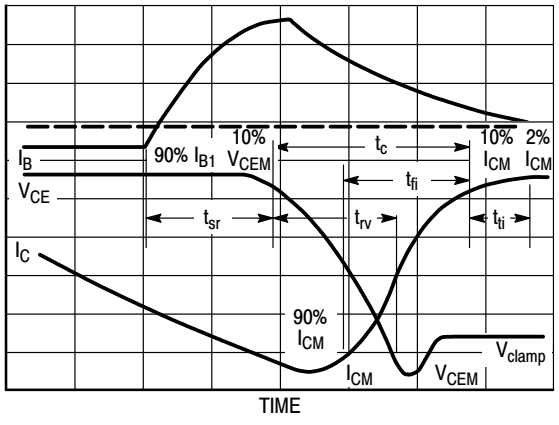


Figure 7. Inductive Switching Measurements

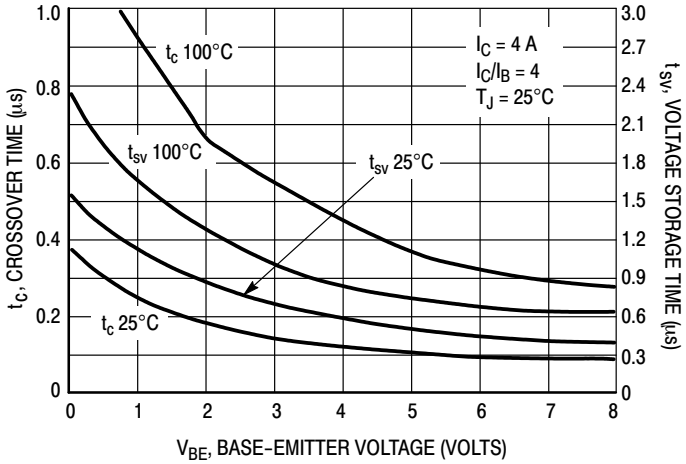


Figure 8. Inductive Switching Times

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SWITCHING TIMES NOTE

In resistive switching circuits, rise, fall, and storage times have been defined and apply to both current and voltage waveforms since they are in phase. However, for inductive loads which are common to SWITCHMODE power supplies and hammer drivers, current and voltage waveforms are not in phase. Therefore, separate measurements must be made on each waveform to determine the total switching time. For this reason, the following new terms have been defined.

t_{sv} = Voltage Storage Time, 90% I_{B1} to 10% V_{CEM}

t_{rv} = Voltage Rise Time, 10−90% V_{CEM}

t_{fi} = Current Fall Time, 90−10% I_{CM}

t_{ti} = Current Tail, 10−2% I_{CM}

t_c = Crossover Time,10% V_{CEM} to 10% I_{CM}

An enlarged portion of the inductive switching waveform is shown in Figure 7 to aid on the visual identity of these terms.

For the designer, there is minimal switching loss during storage time and the predominant switching power losses occur during the crossover interval and can be obtained using the standard equation from AN−222A:

$P_{SWT} = 1/2\ V_{CC} I_C(t_c) f$

In general, $t_{rv} + t_{fi} \simeq t_c$. However, at lower test currents this relationship may not be valid.

As is common with most switching transistors, resistive switching is specified at 25°C and has become a benchmark for designers. However, for designers of high frequency converter circuits, the user oriented specifications which make this a “SWITCHMODE” transistor are the inductive switching speeds (t_c and t_{sv}) which are guaranteed at 100°C.

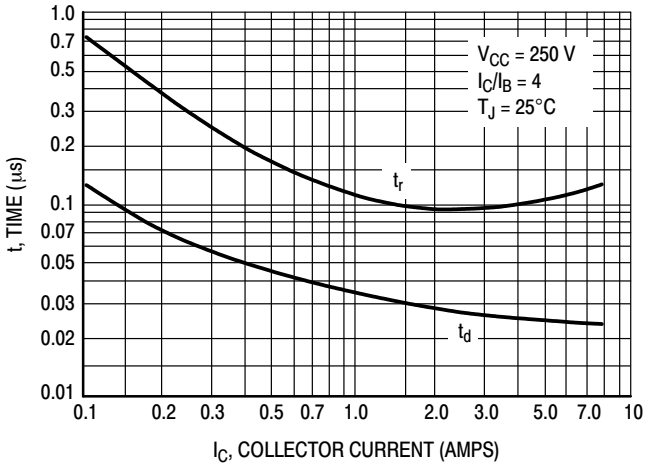


Figure 9. Turn−On Switching Times

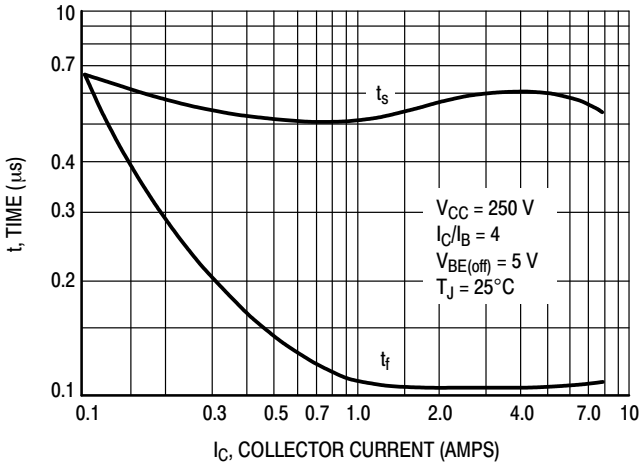


Figure 10. Turn−Off Switching Time

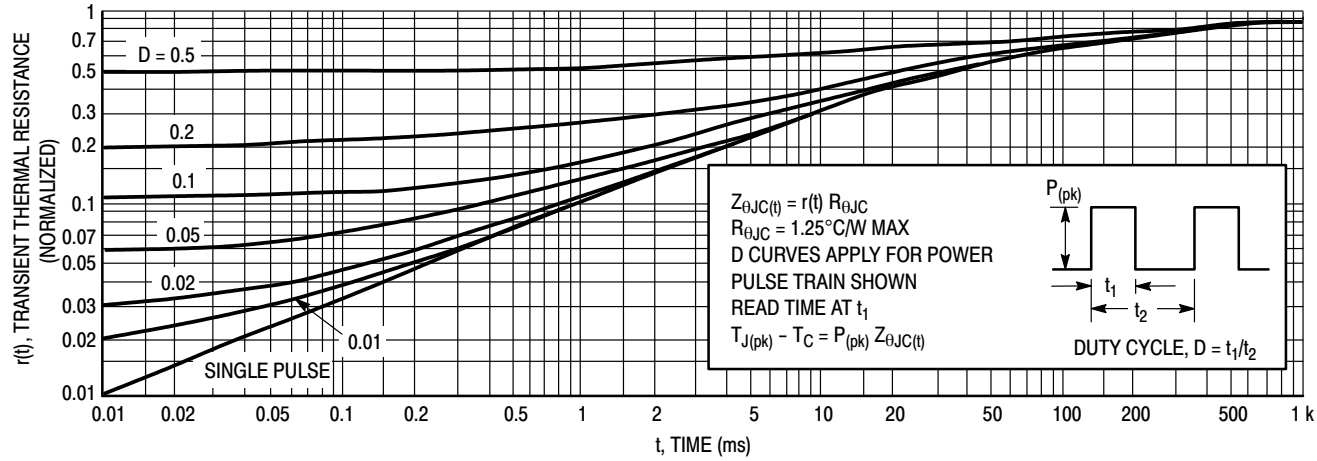


Figure 11. Typical Thermal Response [$Z_{\theta JC}(t)$]

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The Safe Operating Area figures shown in Figures 12 and 13 are specified for these devices under the test conditions shown.

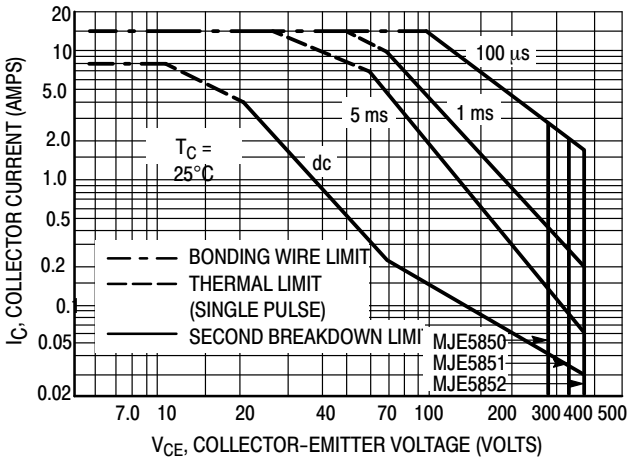


Figure 12. Maximum Forward Bias Safe Operating Area

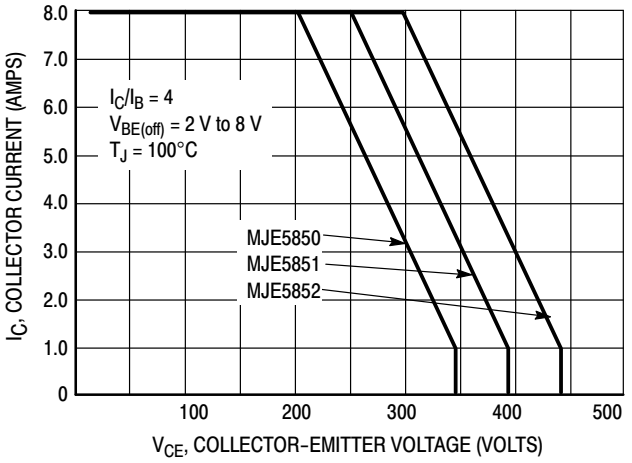


Figure 13. RBSOA, Maximum Reverse Bias Safe Operating Area

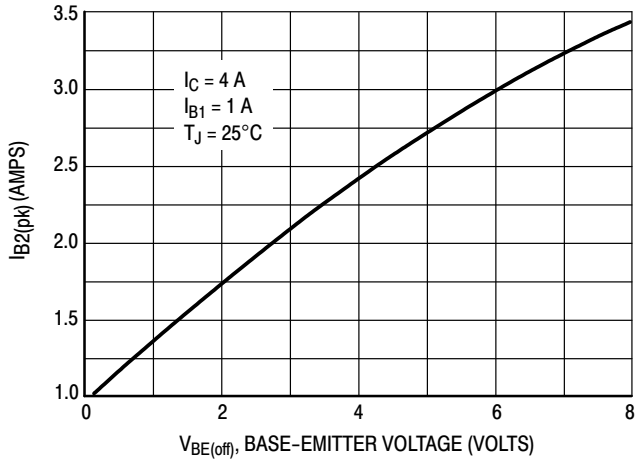


Figure 14. Peak Reverse Base Current

Safe Operating Area Information

Forward Bias

There are two limitations on the power handling ability of a transistor average junction temperature and second breakdown. Safe operating area curves indicate $I_C - V_{CE}$ limits of the transistor that must be observed for reliable operation, i.e., the transistor must not be subjected to greater dissipation than the curves indicate.

The data of Figure 12 is based on $T_C = 25°C$; $T_{J(pk)}$ is variable depending on power level. Second breakdown pulse limits are valid for duty cycles to 10% but must be derated when $T_C \geq 25°C$. Second breakdown limitations do not derate the same as thermal limitations. Allowable current at the voltages shown on Figure 12 may be found at any case temperature by using the appropriate curve on Figure 15.

$T_{J(pk)}$ may be calculated from the data in Figure 11. At high case temperatures, thermal limitations will reduce the power that can be handled to values less than the limitations imposed by second breakdown.

Reverse Bias

For inductive loads, high voltage and high current must be sustained simultaneously during turn−off, in most cases, with the base to emitter junction reverse biased. Under these conditions the collector voltage must be held to a safe level at or below a specific value of collector current. This can be accomplished by several means such as active clamping, RC snubbing, load line shaping, etc. The safe level for these devices is specified as Reverse Bias Safe Operating Area and represents the voltage−current condition allowable during reverse biased turn−off. This rating is verified under clamped conditions so that the device is never subjected to an avalanche mode. Figure 13 gives the RBSOA characteristics.

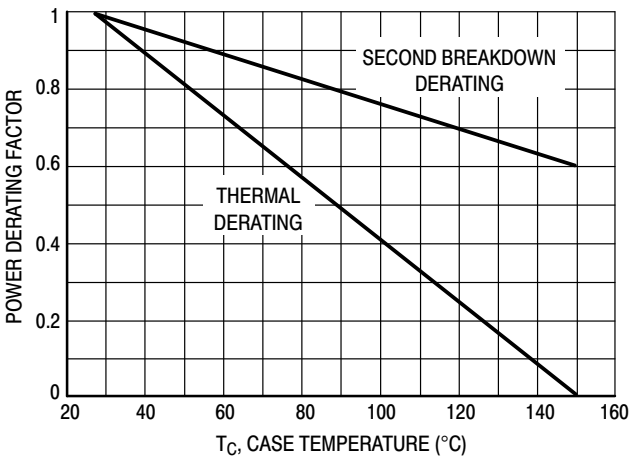


Figure 15. Forward Bias Power Derating

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ORDERING INFORMATION

Device	Package	Shipping
MJE5850G	TO−220 (Pb−Free)	50 Units / Rail
MJE5851G	TO−220 (Pb−Free)	50 Units / Rail

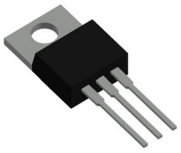
DISCONTINUED (Note 3)

Device	Package	Shipping†
MJE5852G	TO−220 (Pb−Free)	50 Units / Rail

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

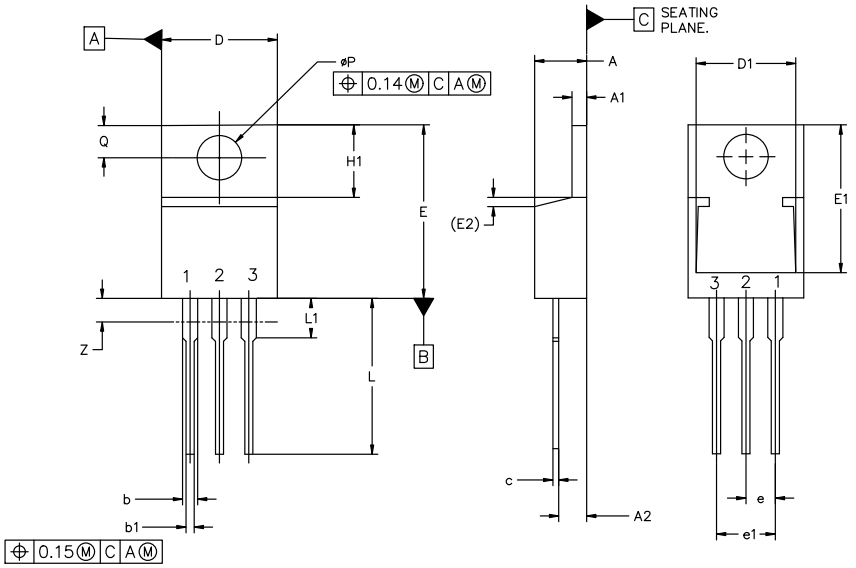
3. **DISCONTINUED:** This device is not recommended for new design. Please contact your **onsemi** representative for information. The most current information on this device may be available on www.onsemi.com.

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TO−220−3 10.10x15.12x4.45, 2.54P
CASE 221A
ISSUE AL

DATE 05 FEB 2025



MILLIMETERS			
DIM	MIN	NOM	MAX
A	4.07	4.45	4.83
A1	1.15	1.28	1.41
A2	2.04	2.42	2.79
b	1.15	1.34	1.52
b1	0.64	0.80	0.96
c	0.36	0.49	0.61
D	9.66	10.10	10.53
D1	8.43	8.63	8.83
E	14.48	15.12	15.75
E1	12.58	12.78	12.98
E2	1.27 REF		

MILLIMETERS			
DIM	MIN	NOM	MAX
e	2.42	2.54	2.66
e1	4.83	5.08	5.33
H1	5.97	6.22	6.47
L	12.70	13.49	14.27
L1	2.80	3.45	4.10
Q	2.54	2.79	3.04
øP	3.60	3.85	4.09
Z	−.−−	−.−−	3.48

NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSION Z DEFINES A ZONE WHERE ALL BODY AND LEAD IRREGULARITIES ARE ALLOWED.

STYLE 1:
PIN 1. BASE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 2:
PIN 1. BASE
2. EMITTER
3. COLLECTOR
4. EMITTER

STYLE 3:
PIN 1. CATHODE
2. ANODE
3. GATE
4. ANODE

STYLE 4:
PIN 1. MAIN TERMINAL 1
2. MAIN TERMINAL 2
3. GATE
4. MAIN TERMINAL 2

STYLE 5:
PIN 1. GATE
2. DRAIN
3. SOURCE
4. DRAIN

STYLE 6:
PIN 1. ANODE
2. CATHODE
3. ANODE
4. CATHODE

STYLE 7:
PIN 1. CATHODE
2. ANODE
3. CATHODE
4. ANODE

STYLE 8:
PIN 1. CATHODE
2. ANODE
3. EXTERNAL TRIP/DELAY
4. ANODE

STYLE 9:
PIN 1. GATE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 10:
PIN 1. GATE
2. SOURCE
3. DRAIN
4. SOURCE

STYLE 11:
PIN 1. DRAIN
2. SOURCE
3. GATE
4. SOURCE

STYLE 12:
PIN 1. MAIN TERMINAL 1
2. MAIN TERMINAL 2
3. GATE
4. NOT CONNECTED

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DESCRIPTION:	**TO−220−3 10.10x15.12x4.45, 2.54P**	**PAGE 1 OF 1**

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